Executive Summary

The Direct Air Capture Blade project, sponsored by San Diego startup Carbon Blade, studied, designed, and fabricated a blade system for atmospheric carbon dioxide removal. Increasing carbon dioxide concentration in the atmosphere is causing widespread adverse effects across the globe. Government bodies and companies, such as Carbon Blade, are working to develop systems that capture and remove carbon dioxide either at the source of emission or from the existing supply in the air (Direct Air Capture, DAC). This project supports Carbon Blade with the latter, specifically on designing a blade system that serves as the surfaces for carbon sequestration. The blade must be coated with a sodium hydroxide solvent, which will react with carbon dioxide passing overhead to produce aqueous sodium bicarbonate solution.

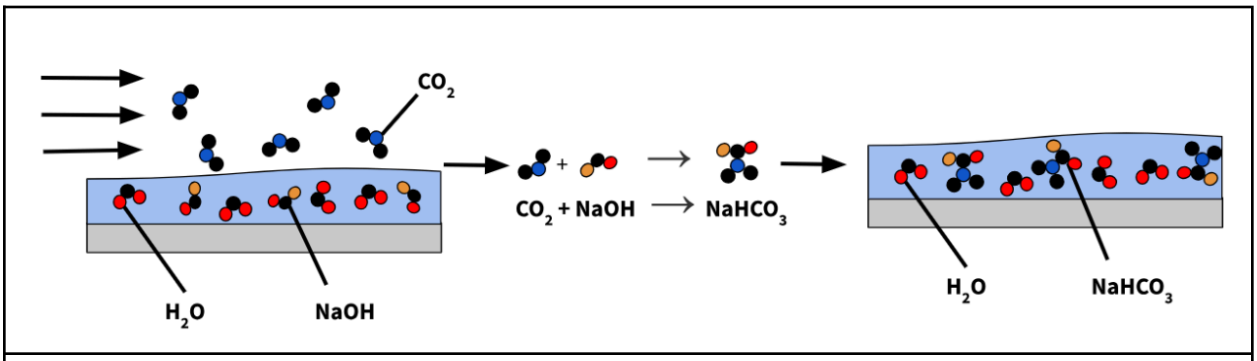

Figure 1. A schematic showing the chemical reaction occurring on the blades.

The blades were designed under the following functional requirements:

- 0.1 metric tons per day of carbon dioxide capture.
- 100 milliliters per minute of solvent flow rate over the blades.
- Capture efficiency of 50% of the passing carbon dioxide.
- 1 rotation per minute rotation rate of the system.
- Minimized power consumption.
- 192m^2 of required surface area for the chemical reaction to occur.

The final design solution that was determined most optimal to work with the functional constraints was a helix where each blade is one sixteenth of a full rotation, and the solvent flow cascading down the blade stack circularly. A total of 100 full rotations will be required to meet the surface area requirements. The blades are spaced apart to allow airflow to push into the gaps and react with all portions of the surface, and the whole system rotates around a central shaft which allows the solvent-covered faces to always be exposed to incoming airflow. One major benefit to this design over other standard flat blade designs lies in the solvent flow between blades. Flat blades require solvent to be collected and redistributed from one story of the blade stack to the next. The helix design allows for the solvent to easily cascade from one blade to another without the need for an external redistribution system. Other key components of the design are the external supports that function to minimize blade deflection. There are 16 supports on the outside edge of the blades, each hold up four blades and are fixed to a rotating disk that transfers the load back to the central shaft. A prototype containing all key systems, the blades,

rotating shaft, external support, baseplate, and solvent delivery/catchment were built for a 4-rotation helix. The blades, solvent containment, external supports, and piping were built from PVC and other structural components were made from wood or sheet metal.

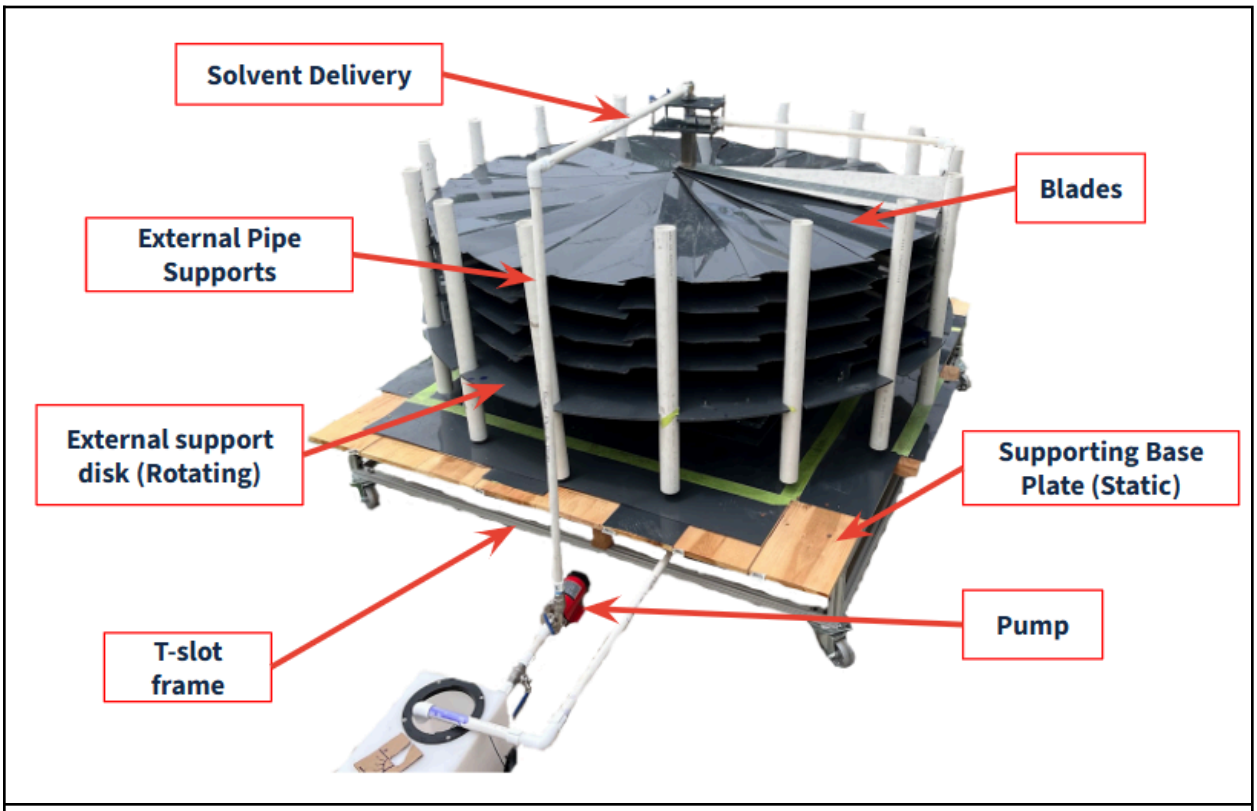

Figure 2. The 4-rotation physical prototype.

CO_2 capture rates are maximized when the solvent covers the entire surface of the blade, air contactors seek to identify the correct blade design, solvent flow rate, rotational rate, or surface treatment to ensure coverage meets this requirement. The helix prototype was tested under varying flow rates (0.88-2.19 GPM) and multiple surface treatment schemes, applying rubbing alcohol and sandpaper to the blades, to determine the maximum surface area coverage. Rubbing alcohol was hypothesized to decrease the contact angle of fluid on the surface, which would increase the footprint of each droplet, and the sandpaper introduced small flow obstructions that forced water to spread orthogonal to the direction of flow. It was found that the maximum surface area coverage, 85.5%, was the result of sandpaper treated blades with the second-highest flow rate (out of four tested rates) of 1.75 GPM. In general, surface treatments significantly increased the solvent surface area coverage, and the highest coverage occurred at 1.75GPM. These results indicate that faster flow rates do not allow the solvent enough time to spread out and cover the blade, while low flow rates do not carry enough momentum for substantial coverage. This prototype serves as a stepping stone to accelerate passive air contactor designs into larger-scale manufacturing.